WINE QUALITY PREDICTION

Problem Statement

Wine quality assessment is an important process in the food and beverage industry, where maintaining consistent product quality is essential for customer satisfaction and market competitiveness. Traditionally, wine quality evaluation is performed through sensory analysis, which can be subjective, time-consuming, and resource-intensive.

This project focuses on developing a machine learning-based predictive system to classify wine quality using physicochemical properties of red Portuguese "Vinho Verde" wine. By analyzing chemical characteristics, the project aims to automate quality prediction and support efficient wine quality control.

Objectives:

- Predict wine quality based on physicochemical features
- Analyze the influence of chemical properties on wine quality
- Build an accurate classification model
- Handle dataset imbalance challenges
- Deploy the predictive model through a user-friendly web application

Dataset Used

Dataset Name:

Red Wine Quality Dataset

Dataset Overview:

- Dataset Type: Classification problem
- Wine Type: Red Portuguese "Vinho Verde" wine
- Target Variable: Quality score (0–10)

Dataset Characteristics:

- Fewer samples
- Highly imbalanced class distribution
- Ordered quality classes
- More normal-quality wines than poor or excellent wines

Input Features:

The dataset includes 11 physicochemical properties:

1. Fixed acidity
2. Volatile acidity
3. Citric acid
4. Residual sugar
5. Chlorides
6. Free sulfur dioxide
7. Total sulfur dioxide
8. Density
9. pH
10. Sulphates
11. Alcohol

Output Variable:

12. Quality

Data Source:

- Kaggle

Approach / Methodology

Data Preparation:

- Loaded the red wine quality dataset
- Selected physicochemical properties as input features
- Used quality score as target variable
- Performed data preprocessing
- Split data into training and testing datasets
- Applied machine learning classification techniques

Machine Learning Model Used:

Random Forest Classifier:

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification performance, reduce overfitting, and enhance predictive accuracy.

Reason for Selection:

- Effective for classification tasks
- Handles feature interactions well
- Reduces overfitting
- Suitable for imbalanced datasets
- Provides strong predictive performance

Web Application Development

Deployment Platform:

Streamlit

Features:

- User-friendly interface
- Allows users to input wine chemical properties
- Predicts wine quality instantly
- Demonstrates real-world practical implementation of the machine learning model

Results and Evaluation

Random Forest Classifier Performance

Accuracy Score:

93.44%

Confusion Matrix:

[[278, 5],
[16, 21]]

Confusion Matrix Interpretation:

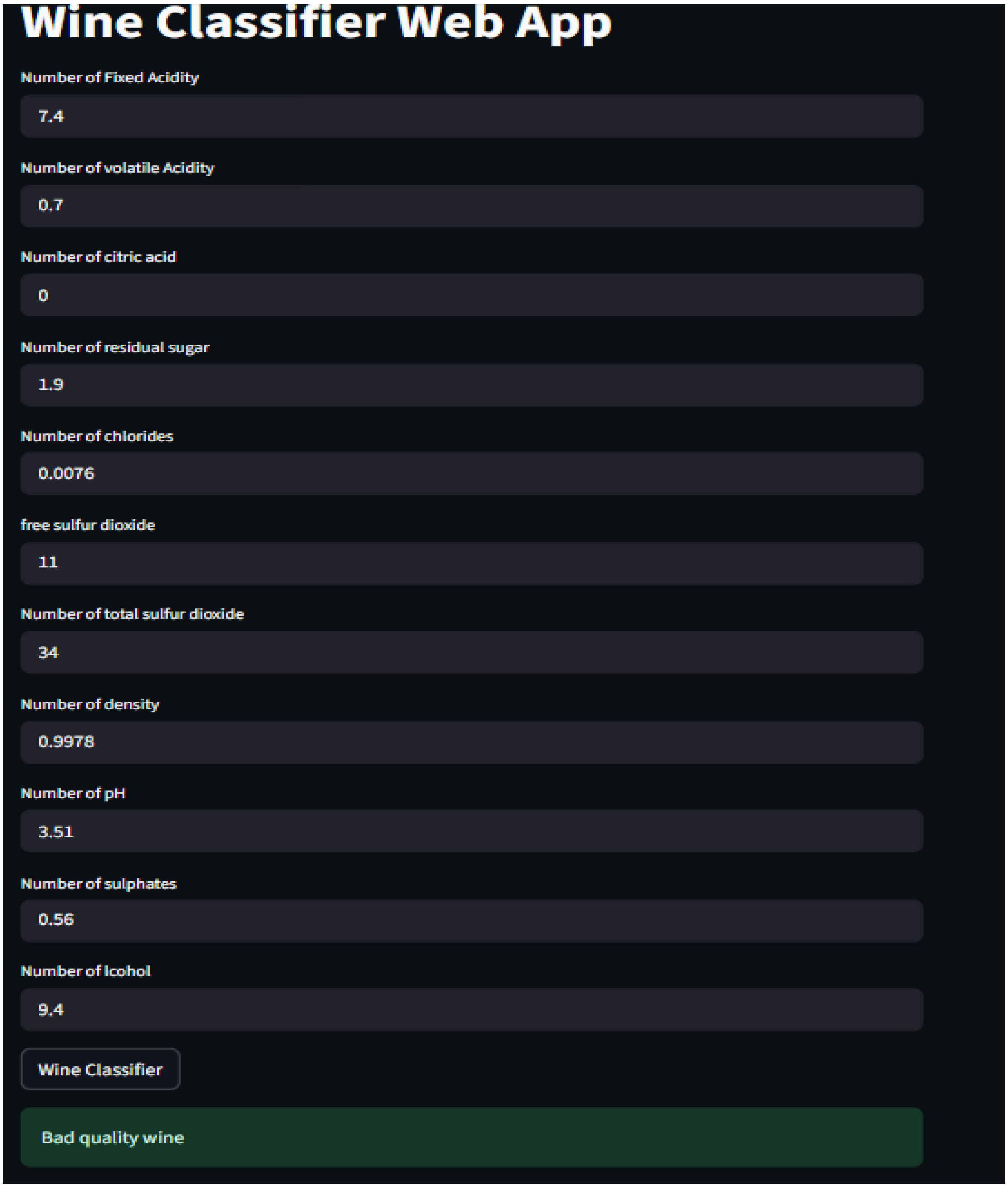
- True Negatives (TN): 278
- False Positives (FP): 5
- False Negatives (FN): 16
- True Positives (TP): 21

Performance Analysis:

- High overall classification accuracy
- Very low false positive rate

- Good prediction of wine quality categories
- Strong generalization despite class imbalance
- Reliable performance for practical deployment

Output



Conclusion

This project successfully developed a machine learning-based wine quality prediction system using physicochemical test data.

Key Achievements:

- Accurate wine quality classification
- Effective handling of dataset complexity
- Streamlit-based deployment for practical usability